



# New User Seminar

**Website:** [www.rc.colorado.edu](http://www.rc.colorado.edu)

**Documentation:** <https://curc.readthedocs.io>

**Helpdesk:** [rc-help@colorado.edu](mailto:rc-help@colorado.edu)



Research Computing  
UNIVERSITY OF COLORADO **BOULDER**



# Meet the User Support Team



Layla  
Freeborn



Brandon  
Reyes



Andy  
Monaghan



Michael  
Schneider



John  
Reiland



Dylan  
Gottlieb



Mohal  
Khandelwal



Ragan  
Lee

# RC Resources

## Tech Support

- High Performance Computing
- Data Management
- Cloud Computing
- Secure Research

## Human Support

- Training Materials & Workshops
- Consultations & Office Hours
- Help Tickets

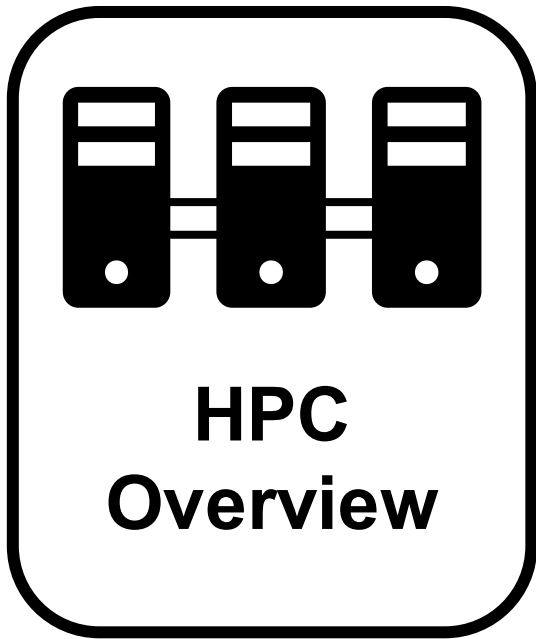


**Ask Questions**

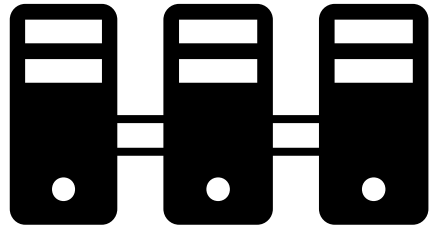


**Discuss Ideas**

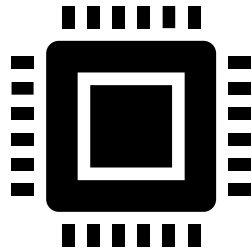
# Learning Goals



# Learning Goals

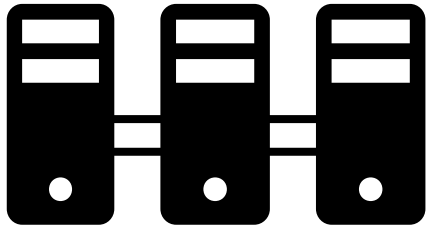


**HPC  
Overview**

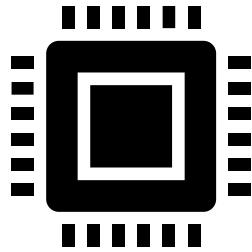


**Nodes &  
Partitions**

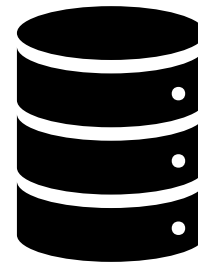
# Learning Goals



**HPC  
Overview**



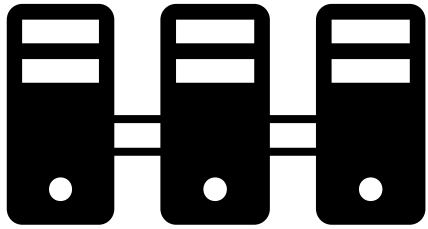
**Nodes &  
Partitions**



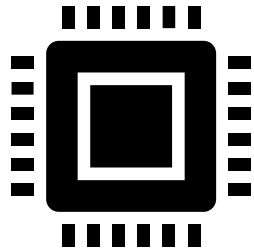
**Data  
Storage**



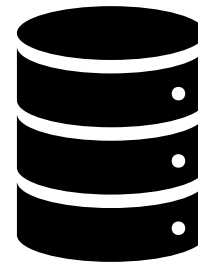
# Learning Goals



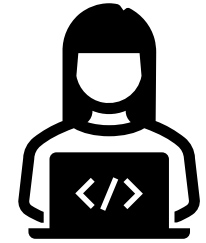
**HPC  
Overview**



**Nodes &  
Partitions**



**Data  
Storage**



**System  
Access**

# High Performance Computing



# High Performance Computing



# High Performance Computing



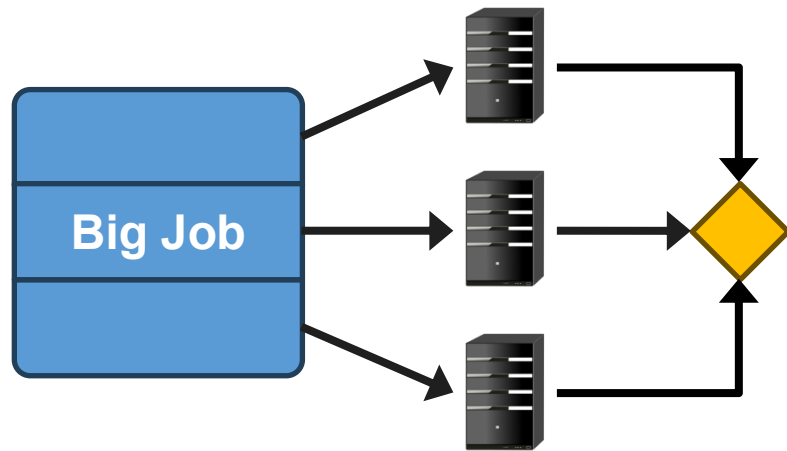
**Scale**

**vs**

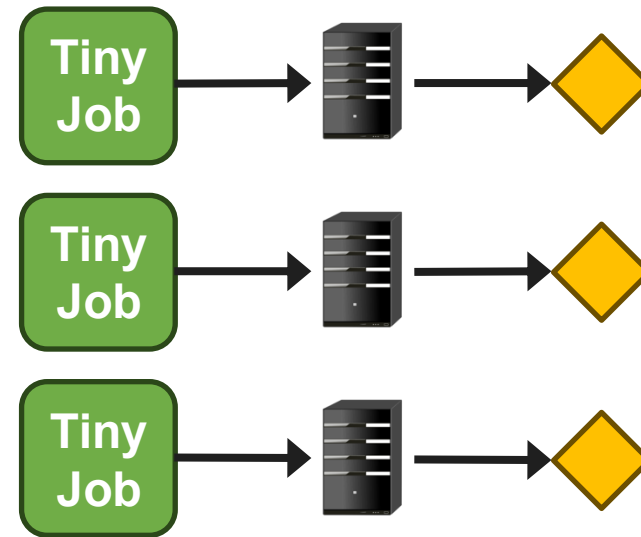
**Speed**



# What can / use HPC for?



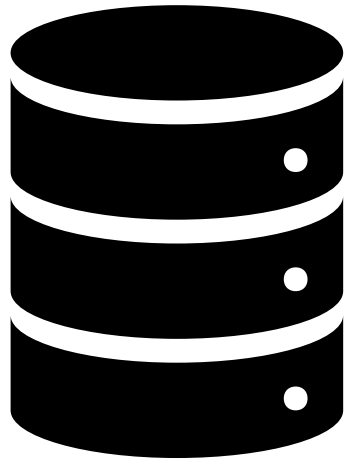
**Parallel Jobs**



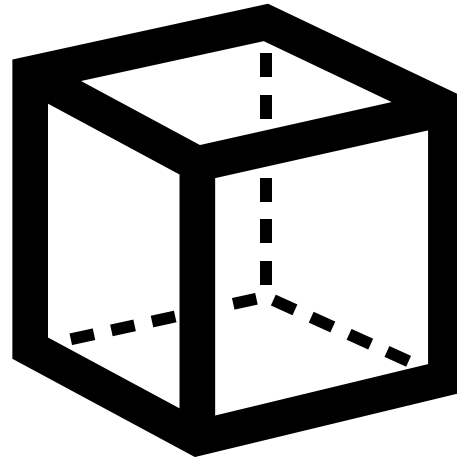
**Serial Jobs**



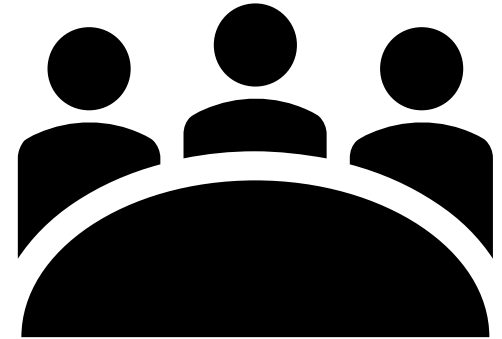
# What can / use HPC for?



**Big Data**



**Viz/Rendering**



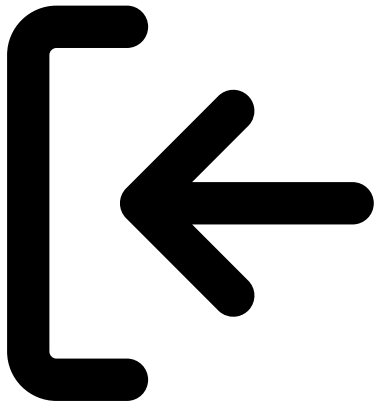
**Shared Work  
Environment**

# HPC Cluster: Alpine

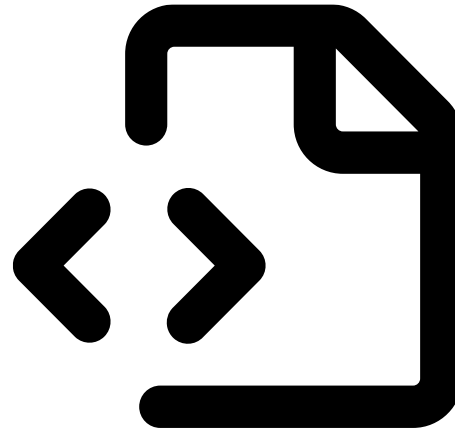


- Heterogeneous cluster
- Hardware provided by CU Boulder, CSU, and AMC
- Access available to CU Boulder, CSU, AMC and RMACC users

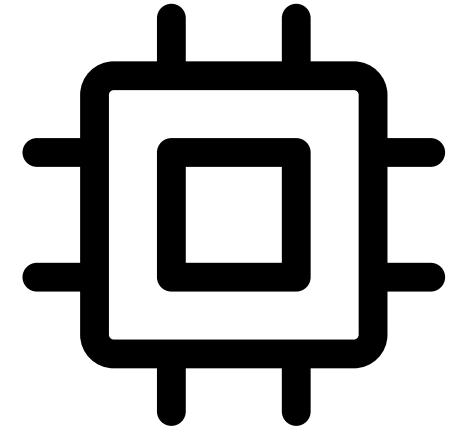
# Node Types



**LOGIN**

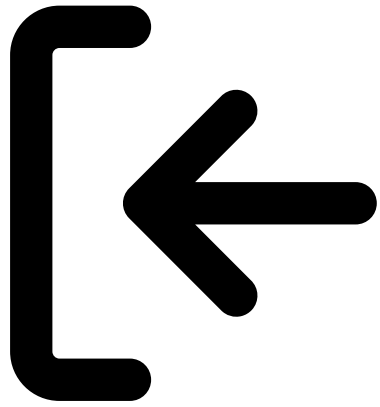


**COMPILE**



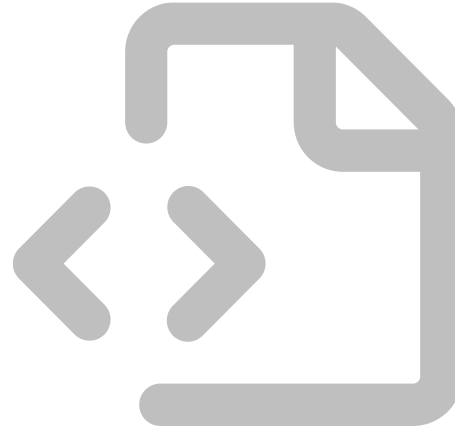
**COMPUTE**

# Node Types

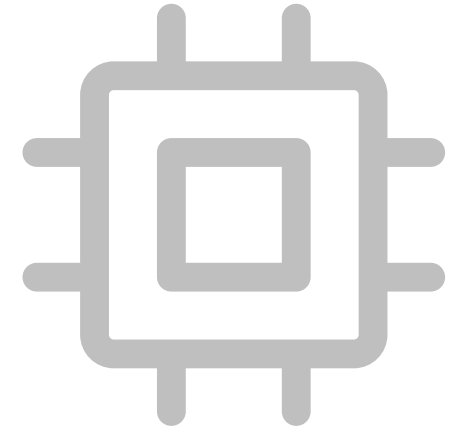


## LOGIN

- Entry to system
- View or edit files
- Submit jobs

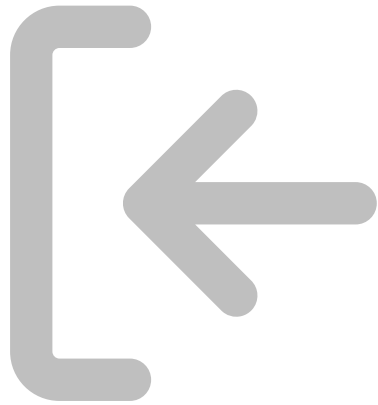


## COMPILE

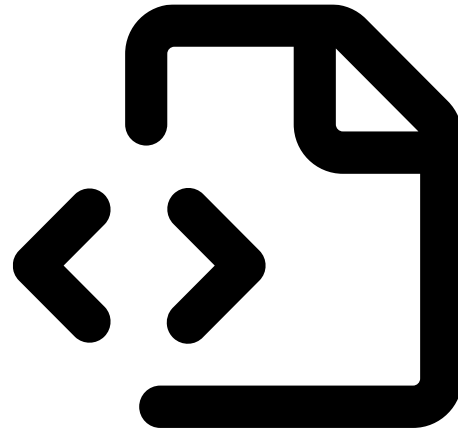


## COMPUTE

# Node Types

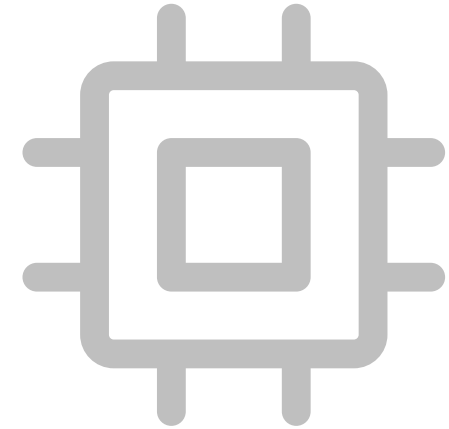


LOGIN



COMPILE

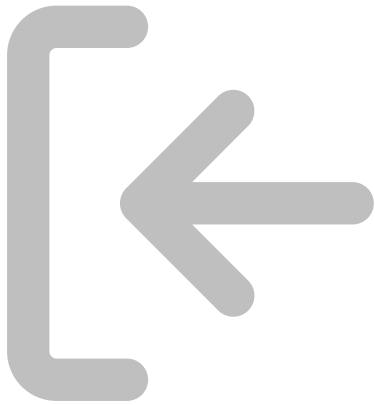
- View or edit files
- Submit jobs
- **Compile code**
- **Install software**



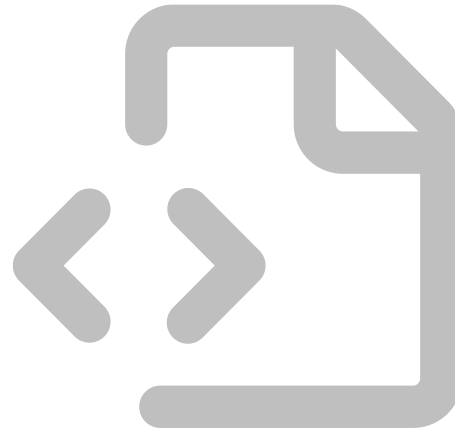
COMPUTE



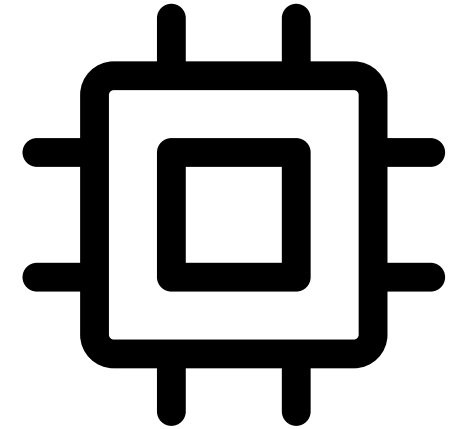
# Node Types



LOGIN



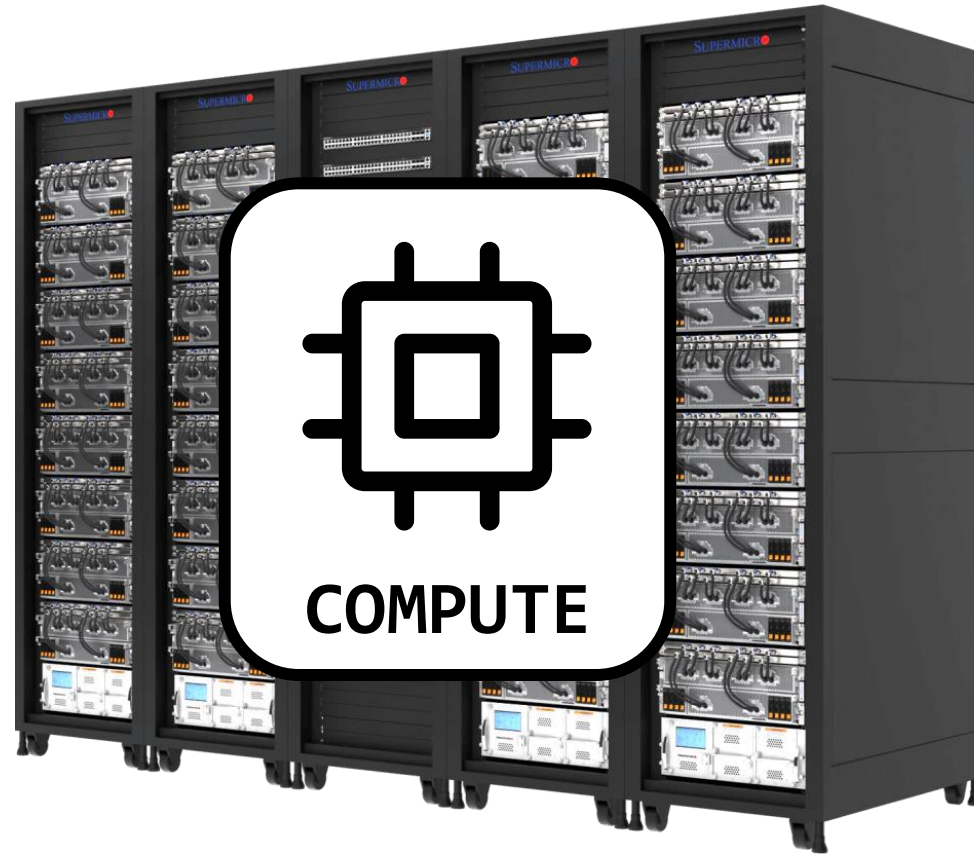
COMPILE



**COMPUTE**

- Run scheduled jobs
- Handle calculations

# Alpine Partitions



# Alpine Partitions

**amilan**  
General Usage



# Alpine Partitions

**amilan**

General Usage

**amem**

High Memory



# Alpine Partitions

**amilan**

General Usage

**amem**

High Memory



**aa100**

Nvidia GPUs

# Alpine Partitions

**amilan**

General Usage

**amem**

High Memory



**aa100**

Nvidia GPUs

**ami100**

AMD GPUs



# Data Storage

## Core

- Personal Storage
- Includes 3 Directories
  - /home (2 GB)
  - /projects (250 GB)
  - /scratch (10 TB)

# Data Storage

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  - /scratch (10 TB)

## PL

- PetaLibrary
- Tiered Storage
  - Active, Archive
- Requires Funding
- Starts at 1 TB

# Core

## Home

- Personal data
- Config files
- OnDemand data
- Do not share!

**2 GB**

# Core

## Home

- Personal data
- Config files
- OnDemand data
- Do not share!

**2 GB**

## Projects

- Code, Job scripts
- Installed software
- Shared data

**250 GB**

# Core

## Home

- Personal data
- Config files
- OnDemand data
- Do not share!

**2 GB**

## Projects

- Code, Job scripts
- Installed software
- Shared data

**250 GB**

## Scratch

- Job data (in/out)
- Shared data
- Deleted after 90 days

**10 TB**

# PetaLibrary

## Active

- Performance tier
- Accessible by all nodes
- No file limit



# PetaLibrary

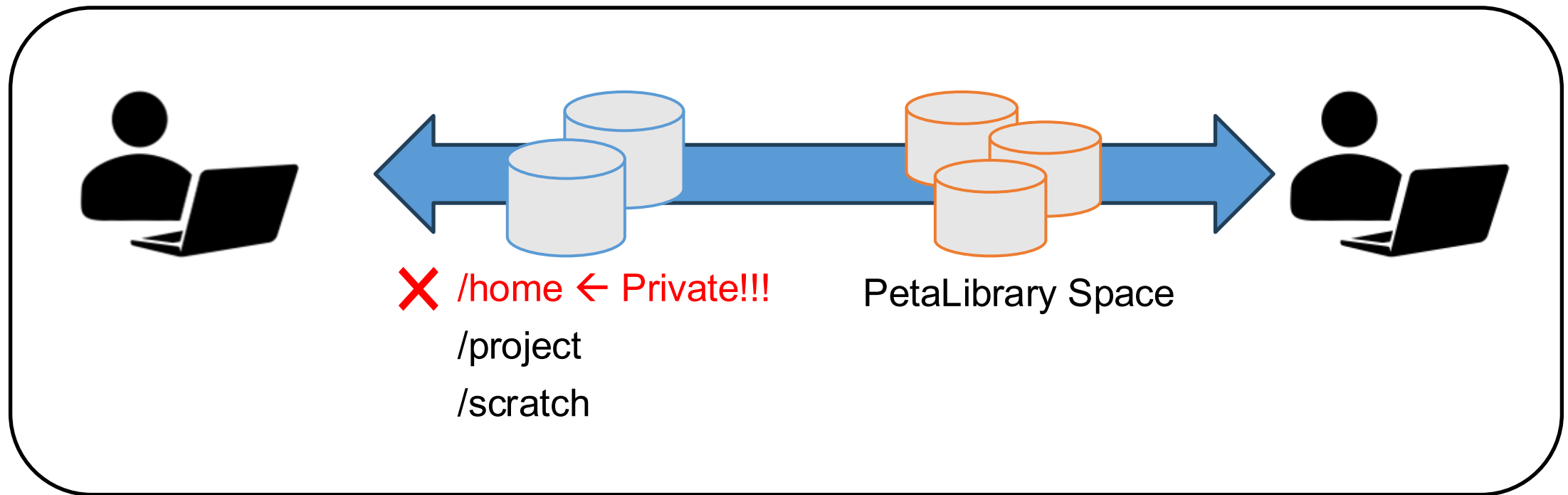
## Active

- Performance tier
- Accessible by all nodes
- No file limit

## Archive

- Integrity tier
- Accessible by login node
- 10,000 file limit

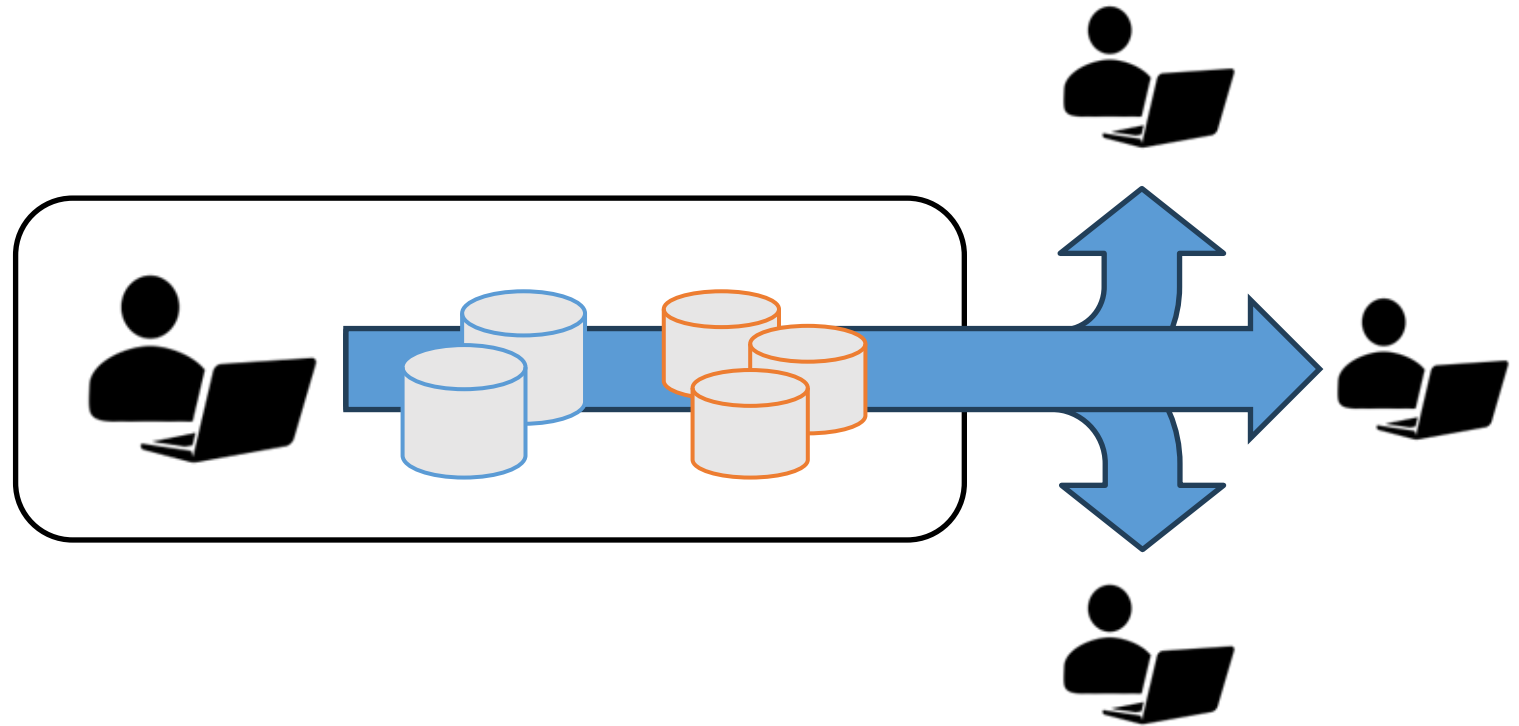
# Data Sharing: Within RC



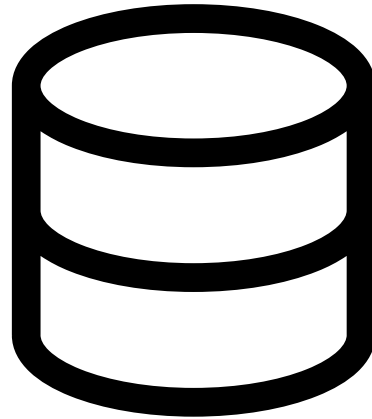
# Data Sharing: Outside RC

## Tools for data transfers:

- Globus (Recommended)
- Terminal/Command Line:
  - rsync
  - rclone
  - sftp
  - scp



# Data Transfer Nodes



## DATA (DTN)

- Support data transfers
- Can be selected when using scp, sftp, or ssh transfers
- *[dtn.rc.colorado.edu](http://dtn.rc.colorado.edu)*

# Acceptable data storage and use

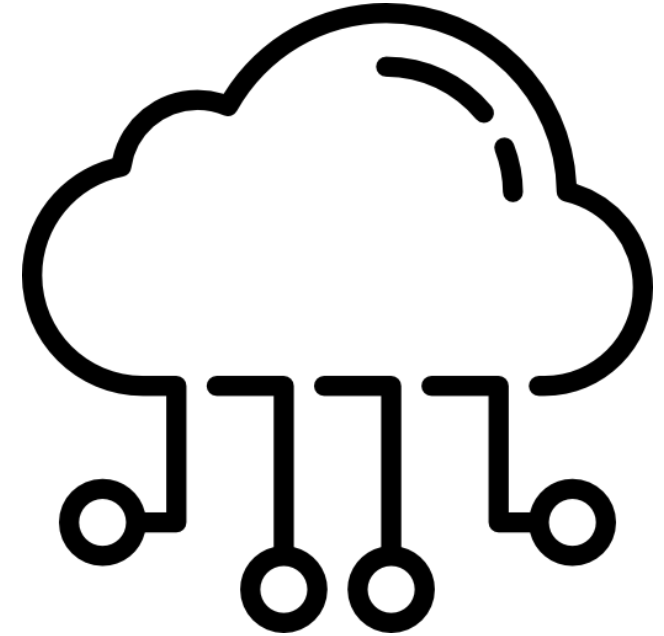
CURC systems and services **should not be used to store** any data that is US government Classified, nor any Controlled Unclassified Information.

For users requiring storage for sensitive data types, please see the secure research computing resources:

<https://www.colorado.edu/rc/secure-research-computing-resources>

# Cloud Computing

- CURC supports AWS, Azure, and GCP
- Alternative or augmentation to HPC
- Interested? Contact [rc-help@colorado.edu](mailto:rc-help@colorado.edu)



# How to Access RC Resources?

1. Get an RC account
2. Set up two-factor authentication with Duo
3. Log in via 1) *terminal* or 2) *OnDemand*
4. Create greatness! (responsibly)



# Getting an Account

- CU Boulder, CSU users and affiliates:
  - Request an account through the RC Account request portal:  
<https://rcamp.rc.colorado.edu/accounts/account-request/create/organization>
- AMC, RMACC users and affiliates:
  - Visit the *AMC* or *RMACC* tab under “Getting a CURC account” in the CURC documentation:  
[https://curc.readthedocs.io/en/latest/getting\\_started/logging-in.html](https://curc.readthedocs.io/en/latest/getting_started/logging-in.html)

# Your RC Account

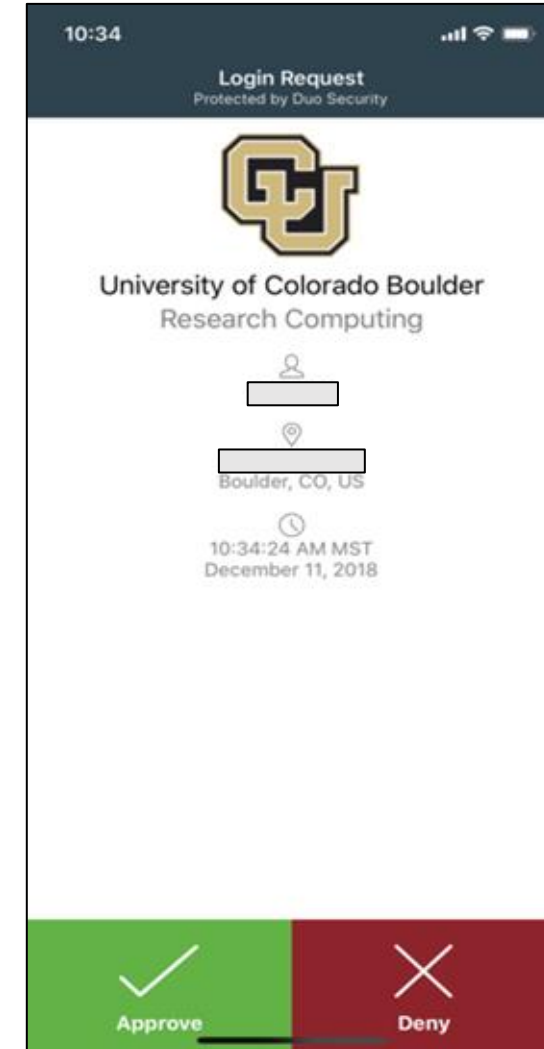
## Access to:

1. Alpine Cluster
2. Core Storage
3. PetaLibrary Storage\*
4. OnDemand
5. Approximately 2,000 Service Units (SUs) per month

*\*If purchased*

# Duo Authentication

1. Duo smartphone app (recommended)
2. Phone Call/Text is an alternatives



# Terminal Access

- Mac or Linux
  - Terminal application
- Windows
  - PuTTY
  - Powershell

```
[user0083@tlogin1 ~]$ pwd  
/home/user0083  
[user0083@tlogin1 ~]$ █
```

# Demo: Logging in via Terminal

To login to an RC login node<sup>\*</sup> :

```
$ ssh <username>@login.rc.colorado.edu
```

Supply your institutional password and your Duo app will alert you to confirm the login

*\* Exclusive to CU and CSU accounts. Available on request for AMC. Not available to RMACC.*

# Demo: logging in with OnDemand

CURC OnDemand is a browser based, integrated, single access point for all of your HPC resources at CU Research Computing.

- CU Boulder: Visit <https://ondemand.rc.colorado.edu>.
- All other Institutions: Visit <https://ondemand-rmacc.rc.colorado.edu/>

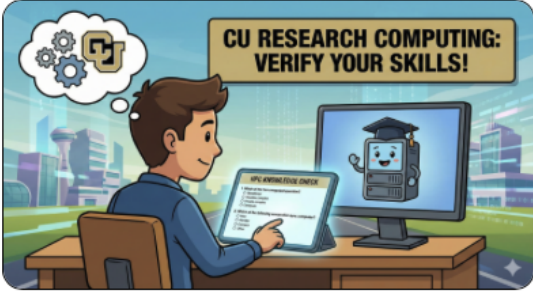
# Help! I'm stuck, where do I go?

- **Documentation**: [curc.readthedocs.io/](http://curc.readthedocs.io/)
- **Trainings with Center for Research Data and Digital Scholarship (CRDDS)**: <https://www.colorado.edu/crdds/>
- **Helpdesk**: [rc-help@colorado.edu](mailto:rc-help@colorado.edu)
- **Consult Hours** (Tuesday 12:00-1:00 in-person or virtually, Thursday 1:00-2:00 virtually)
- Or....

# ...try CURC's new self-paced training!

## Research Computing New User Training

Welcome to the Research Computing New User Training! This course covers the ins and outs of CU Research Computing's (CURC) High Performance Computing (HPC) systems.



Please enter your Research Computing account information:

Do you have an existing RC account?

Select Yes or No

Institutional email address:

Enter institutional email

Submit

**Note:** Multiple questions are embedded in this training. It is ok if you don't know the answer to every question! Many of the questions are designed to test for common misconceptions and help you avoid common pitfalls for new users.

[https://researchcomputing.github.io/New\\_User\\_Async\\_Training](https://researchcomputing.github.io/New_User_Async_Training)



# Helpdesk Tickets

To: [rc-help@colorado.edu](mailto:rc-help@colorado.edu)

Dear Research Computing,

Help! My code won't run!  
Help!

Help please,  
John

To: [rc-help@colorado.edu](mailto:rc-help@colorado.edu)

Dear Research Computing,

I am running into issues running my Python script. I am using a conda environment called my\_python\_env with the pytorch software, and I am receiving the following error. I am not sure how to troubleshoot. My job ID is 620350. Let me know what I can try!

sbatch: error: Batch job submission failed: Invalid partition name specified.

Thanks,  
John

# Survey and feedback

<http://tinyurl.com/curc-survey18>

